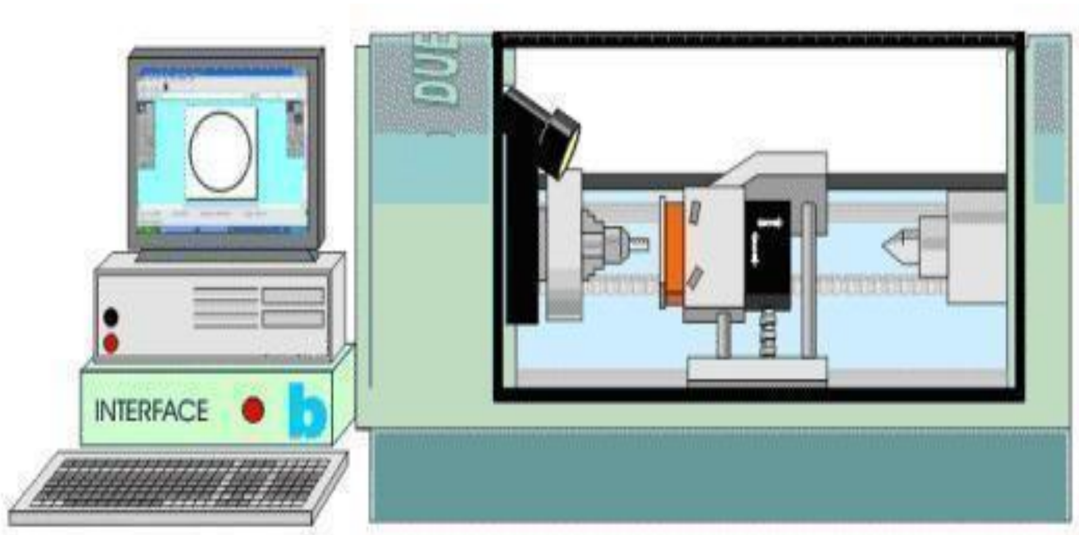
# INTRODUCTION

In Industry it is not efficient or profitable to make everyday products by hand. On a CNC machine it is possible to make hundreds or even thousands of the same items in a day. First a design is drawn using design software, and then it is processed by the computer and manufactured using the CNC machine. This is a small CNC machine and can be used to machine woods, plastics and aluminum. In industry, CNC machines can be extremely large.



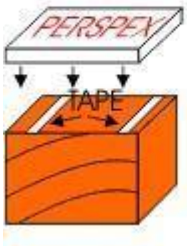
## MEANING OF 'CNC'

CNC means Computer Numerical Control. This means a **computer** converts the design into **numbers** which the computer uses to **control** the cutting and shaping of the material.

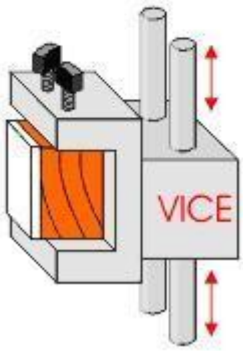

## USE OF TYPICAL CNC MACHINE

1. The design is loaded into the computer which is attached to the CNC machine. The computer changes the design into a special code (numerical) that controls the way the CNC cuts and shapes the material.

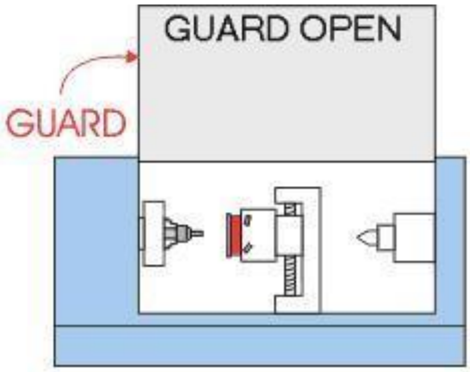
The material to be shaped is taped on to a block with double sided tape. This must be done carefully so that it does not come off the block during machining.




1. The block is then placed in the vice, inside the CNC. It must be tightened up carefully. If it is not secure when the machine starts to cut the material it can come away from the vice. When the machine starts working, the vice moves up, down, right and left according to the design.

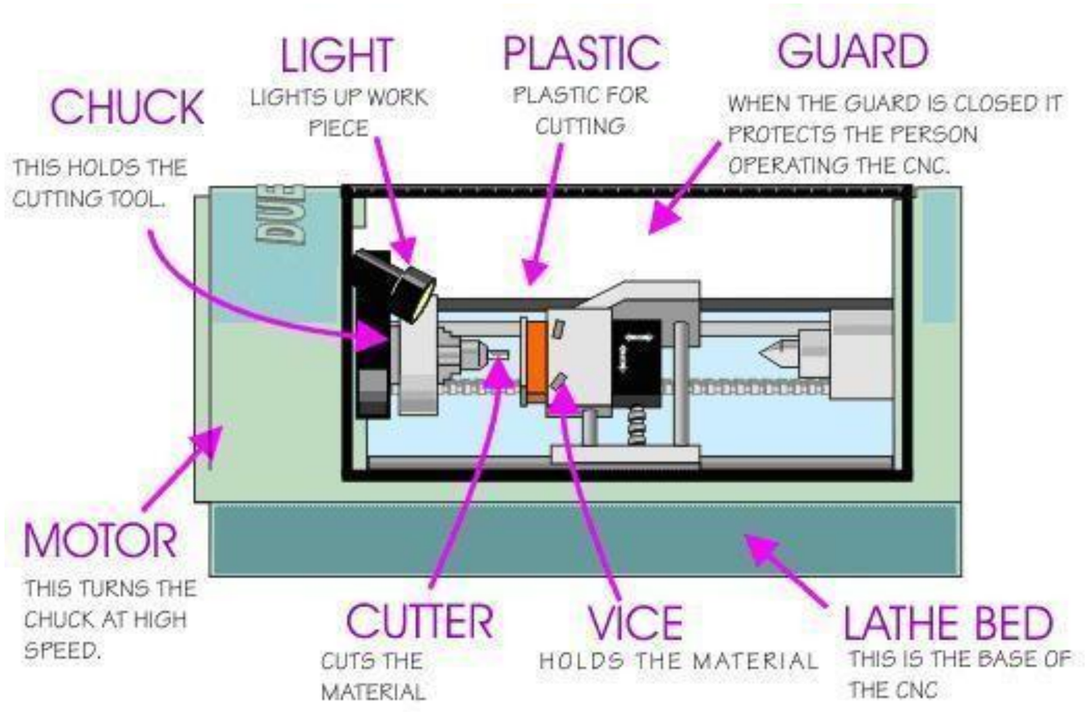



2. The guard is placed in position. It protects the machine operator in case the material is pulled out of the vice by the power of the cutter. For safety reasons, if the guard is not in position the motor will not start.




3. The CNC is turned on and the shape is cut from the material. When the cutter has stopped the shaped material can be removed from the vice.

## PARTS OF CNC MACHINE




**VICE*:*** This holds the material to be cut or shaped. Material must be held securely otherwise it may 'fly' out of the vice when the CNC begins to machine. Normally the vice will be like a clamp that holds the material in the correct position.

**GUARD*:*** The guard protects the person using the CNC. When the CNC is machining the material small pieces can be 'shoot' off the material at high speed. This could be dangerous if a piece hit the person operating the machine. The guard completely encloses the dangerous areas of the CNC.

**CHUCK:** This holds the material that is to be shaped. The material must be placed in it very carefully so that when the CNC is working the material is not thrown out at high speed.

**MOTOR*:*** The motor is enclosed inside the machine. This is the part that rotates the chuck at high speed.

**LATHE BED*:*** The base of the machine. Usually a CNC is bolted down so that it cannot move through the vibration of the machine when it is working.

**CUTTING TOOL*:*** This is usually made from high quality steel and it is the part that actually cuts the material to be shaped.

## CNC MACHINE - INPUT, PROCESS, OUTPUT

A CNC production facility needs three pieces of equipment:

A Computer:

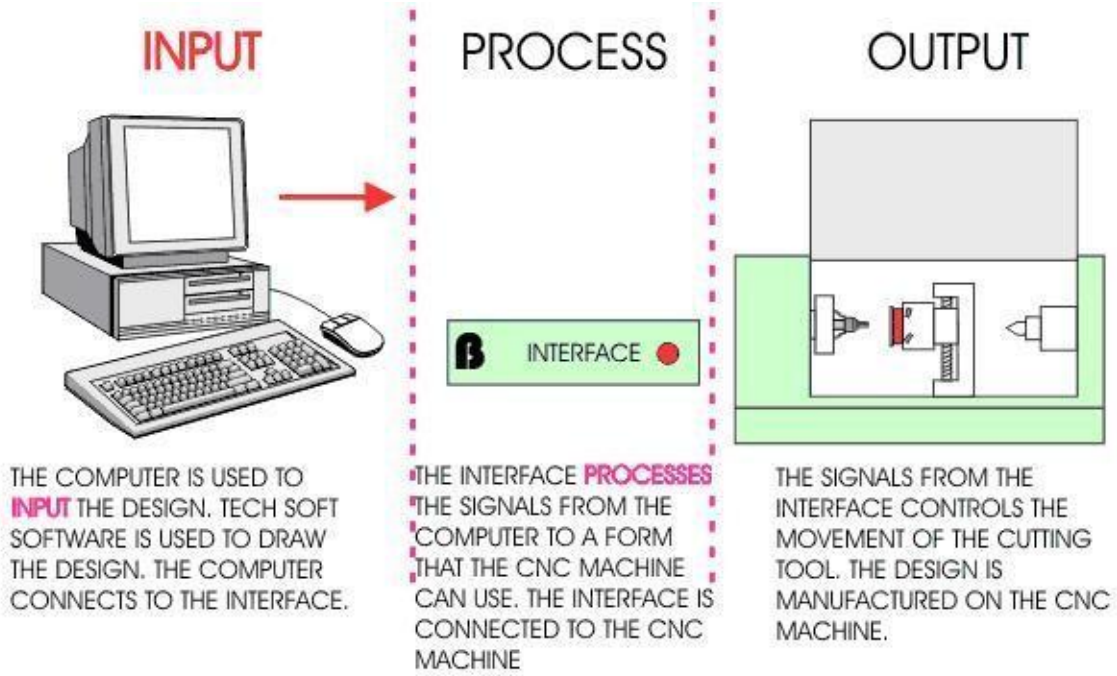
The computer is used to draw the design. However, the design is only a picture and the CNC machine cannot use this to manufacture the product. The computer software must also convert the drawing into numbers (coordinates) that the CNC machine can use when it starts to cut and shape the material.

An Interface:

A computer cannot be directly connected to a CNC machine. The computer is connected to an interface. This converts the signals from the computer to a form that the CNC machine understands. The signals are in the form of digital signals when they are sent to the CNC machine.

CNC (Computer Numerical Control) Machine:

The signals from the interface control the motors on the CNC machine. The signals determine the way the vice moves. The vice moves in three directions X, Y and Z. (Horizontally, vertically and depth). The signals also control the speed of the cutting tool.




## CONTROL PANEL OF CNC

A CNC machine is normally controlled by a computer and software. However, most CNC machines have a range of controls for manual use. It is rare for a CNC machine to be used manually as simple operations are best carried out on cheap/basic/manual machines. When a CNC machine is used manually it is been used well below its capability and specification.

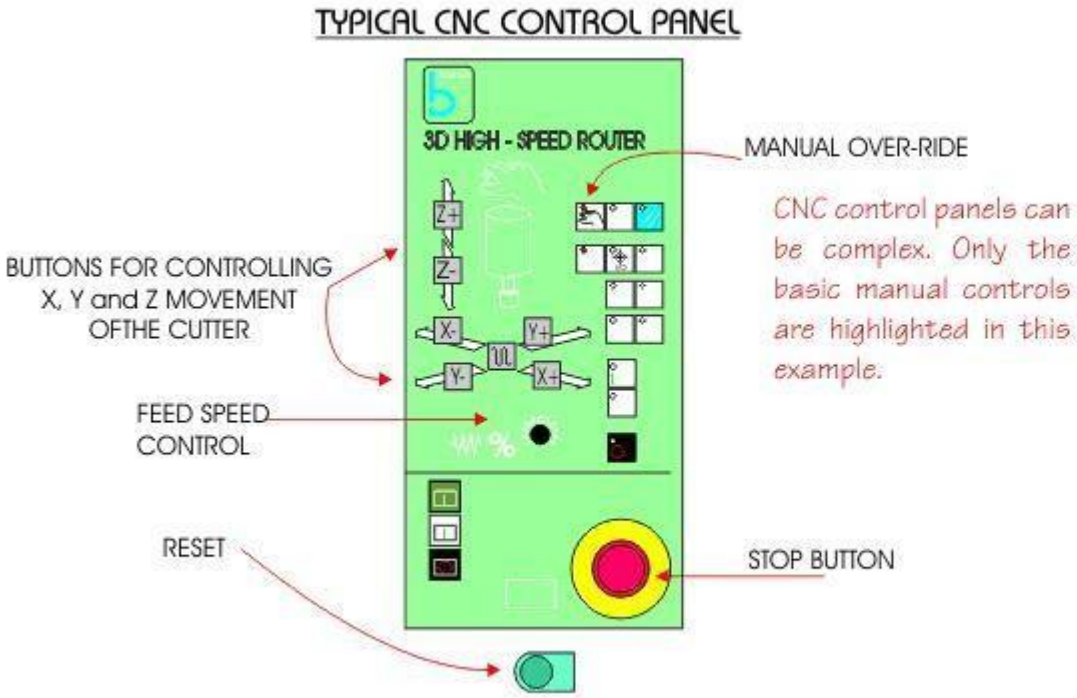
**RESET BUTTON**: The most important control button is usually the reset button. When the CNC machine is turned on, the reset button is pressed by the machine operator. This 'zeros' the cutter, moving the cutter to coordinates 0, 0, 0 on the X, And Z axis. In simple terms, the reset button moves the cutter to the corner of the machine, above the work table.

If the reset button is not pressed, it is possible that the CNC machine will start cutting the material in the wrong place or even miss cutting the material and plunge into the work table.

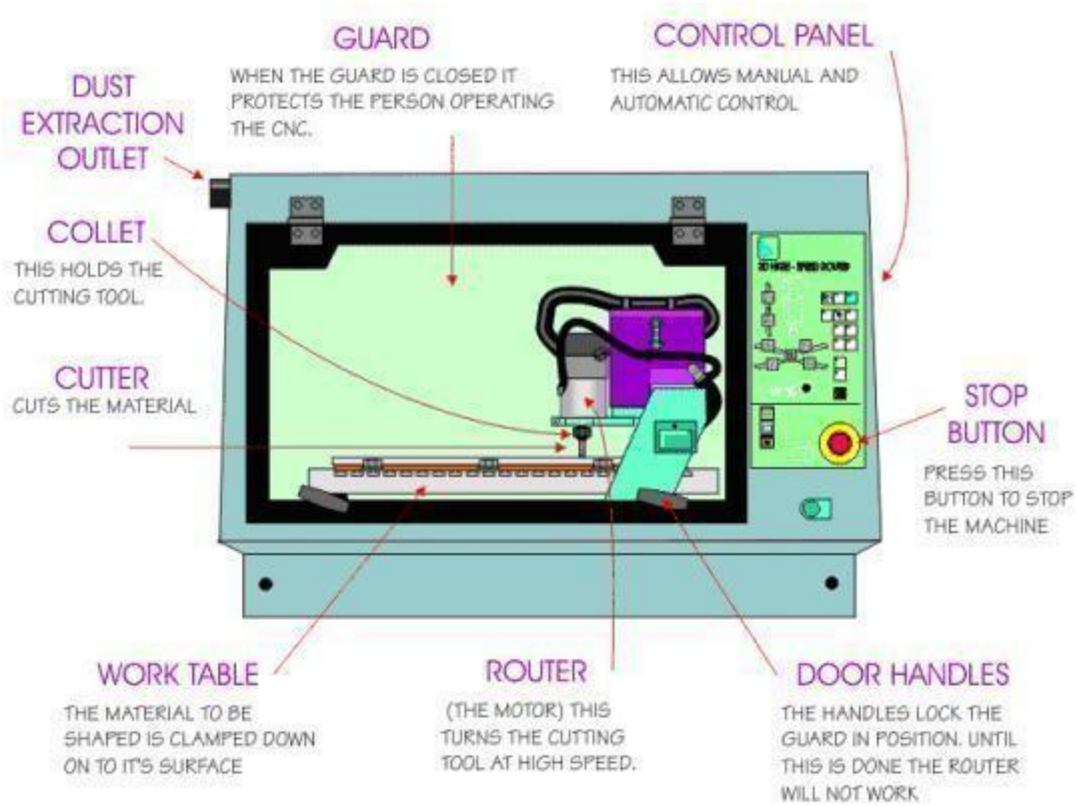
**MANUAL CONTROL**: The cutter can be controlled manually although this is rarely needed. The 'X' and 'Y' buttons control the movement of the cutter along the horizontal surfaces. The 'Z' buttons control depth and up / down movement.

**STOP BUTTON**: Most control panels have stop buttons. When pressed these stop the machine very quickly.

**SPEED AND FEED**: On some CNC machines it is possible to manually vary the speed and feed of the cutter.

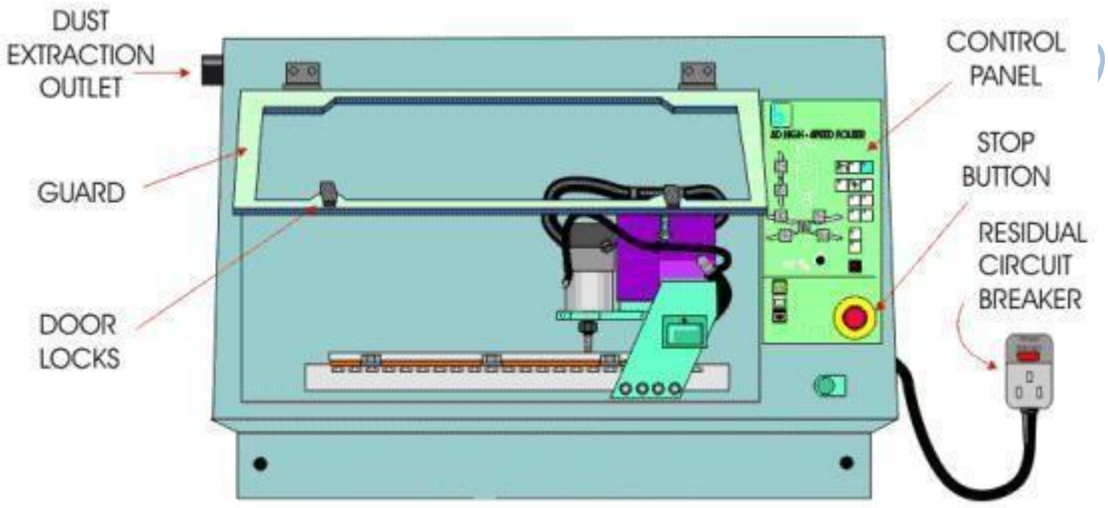

TYPICAL CNC CONTROL PANEL
3D HIGH - SPEED ROUTER
MANUAL OVER-RIDE
CNC control panels can be complex. Only the basic manual controls are highlighted in this example.
BUTTONS FOR CONTROLLING X, Y and Z MOVEMENT OFTHE CUTTER
Z+
Z-
X-
Y+
Y-
X+
FEED SPEED CONTROL
%
RESET
STOP BUTTON

# CNC ROUTER - THE IMPORTANT PARTS



# CNC MACHINES AND SAFETY

CNC machines are very safe to use as they are designed to be as safe as possible. One of the main advantages of CNC machines is that they are much safer than manually operated machines.




1. Most modern CNC machines are designed so that the cutting tool will not start unless the guard is in position. Also, the best CNC machines automatically lock the guard in position whilst the cutter is shaping material. The guard can only be opened if the cutter has stopped.

2. It is essential that pupils / students / machine operators receive quality' instruction before attempting to use any CNC equipment.

3. CNC routers, used for shaping materials such as woods and plastics, have built in extraction. Dust can be very dangerous if inhaled and can also cause eye irritation. The CNC Router shown above has an outlet for an extraction unit. As the router is fully enclosed, dust cannot escape into the atmosphere. If an extraction unit is attached the dust is removed automatically. Most manually operated machine routers

have very limited extraction systems which leave some dust in the air.

4. The CNC router above has a single phase electrical supply. Older machines such as manually operated milling machines and centre lathes have three phase supplies. A single phase electrical supply can be plugged’ into any available socket. The electrical supply for the machine comes through a residual circuit breaker (RCB). If an electrical fault develops the RCB will cut off electrical power immediately.

5. Single phase CNC machines can be moved more easily because they are simply unplugged and relocated. Three phase machines are specially wired by an electrician into the electrical supply and cannot be unplugged.

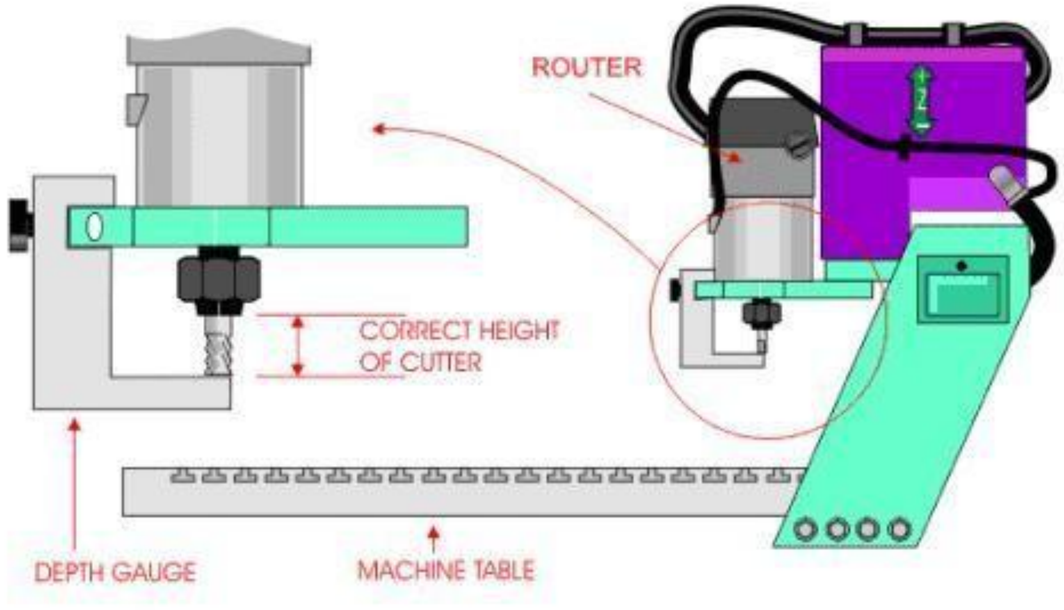
6. Most CNC machines work behind a guard or even a closed, transparent safety door. This means that the operated cannot be hurt by 'flying' pieces of sharp/hot material.

7. Commonsense applies to the use of all machines including CNC machines. Basic safety training regarding working in a workshop and with other machines applies to CNC machines as well.

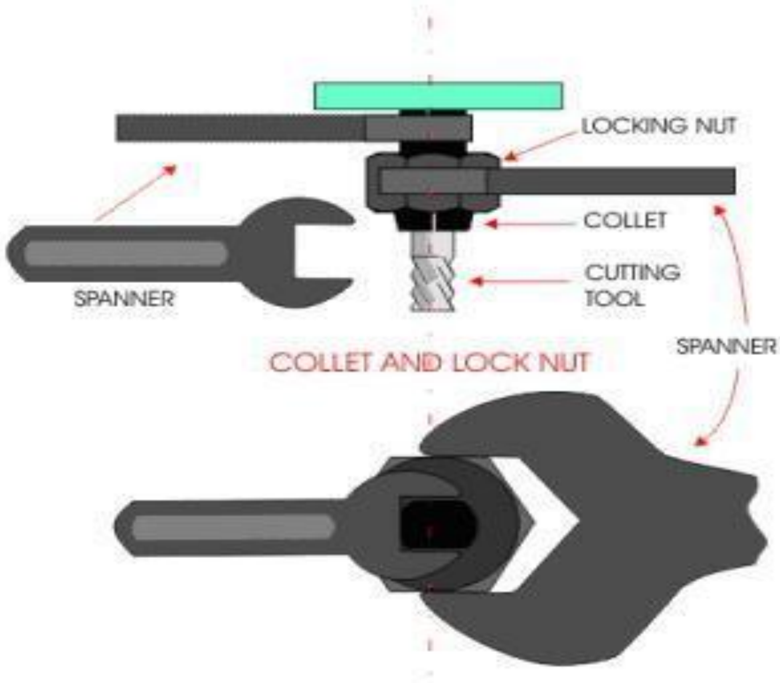
# SETTING UP THE CUTTING TOOL TO THECORRECT LENGTH

One of the few operations that the machine operator carries out is to change the cutting tool. Each CNC machine has a range of cutting tools. Straight cutters chamfer V-groove and radius cutters are some examples. If a detailed design is being manufactured, it may be necessary to change the cutting tool at least once during the manufacturing process. It is very important that all the cutters are set up to exactly to the same length in the cullet. If this is not done the material being machined will be machined at incorrect depths.

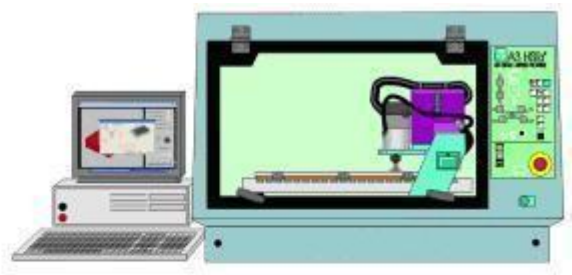
A special depth gauge is used to accurately set up the cutting tools. (See diagram below). This 'rule' applies to all CNC machines although different techniques may be used depending on the type of the CNC machine.



A pair of spanners is used to loosen the cullet and locking nut. The cutting tool can then be removed and the new tool put in position. Once the depth gauge has be used to check the distance from the end of the cutting tool to the cullet, the spanners are used again to tighten the cullet and locking nut.



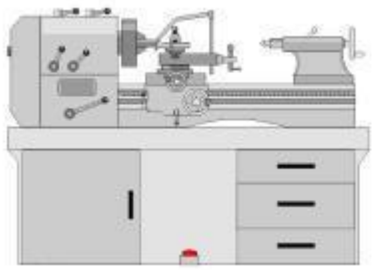
# ADVANTAGES AND DISADVANTAGESOF CNC MACHINES




COMPUTER NUMERICAL CONTROL MACHINE

## ADVANTAGES

1. CNC machines can be used continuously 24 hours a day, 365 days a year and only need to be switched off for occasional maintenance.
2. CNC machines are programmed with a design which can then be manufactured hundreds or even thousands of times. Each manufactured product will be exactly the same.
3. Less skilled/trained people can operate CNC's unlike manual lathes/ milling machines etc... Which need skilled engineers?
4. CNC machines can be updated by improving the software used to drive the machines



MANUALLY OPERATED CENTRE LATHE

## DISADVANTAGES

1. CNC machines are more expensive than manually operated machines, although costs are slowly coming down.
2. The CNC machine operator only needs basic training and skills, enough to supervise several machines. In years gone by, engineers needed years of training to operate centre lathes, milling machines and other manually operated machines. This means many of the old skills are been lost.
3. Less workers are required to operate CNC machines compared to manually operated machines.

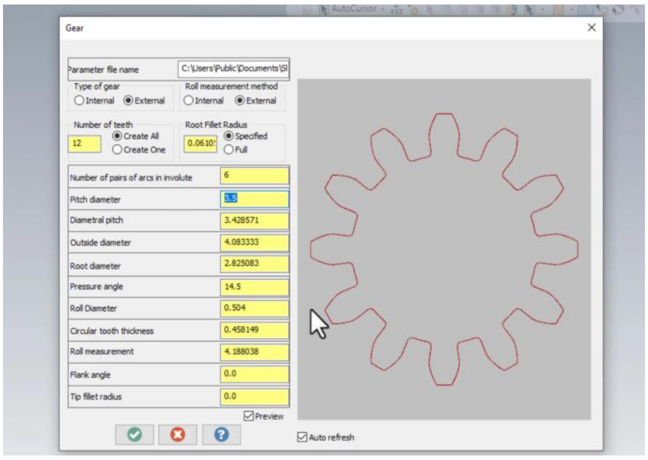
Investment in CNC machines can

| | | |
|---|---|---|
| 5. CNC machines can be used continuously 24 hours a day, 365 days a year and only need to be switched off for occasional maintenance.<br>6. CNC machines are programmed with a design which can then be manufactured hundreds or even thousands of times. Each manufactured product will be exactly the same.<br>7. Less skilled/trained people can operate CNC's unlike manual lathes/ milling machines etc... Which need skilled engineers?<br>8. CNC machines can be updated by improving the software used to drive the machines | | 4. CNC machines are more expensive than manually operated machines, although costs are slowly coming down.<br>5. The CNC machine operator only needs basic training and skills, enough to supervise several machines. In years gone by, engineers needed years of training to operate centre lathes, milling machines and other manually operated machines. This means many of the old skills are been lost.<br>6. Less workers are required to operate CNC machines compared to manually operated machines.<br>Investment in CNC machines can |

# 2D COMPUTER AIDED DESIGN AND MANUFACTURE

There are two types of computer aided design software. 2D design software allows the designer to design shapes with very limited three dimensional properties. Do not underestimate the designs that can be achieved through 2D software.

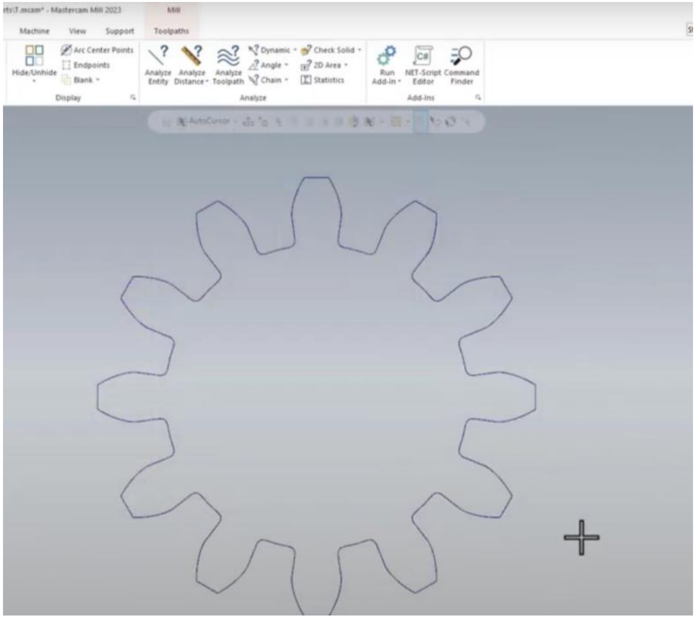
1. The design is drawn using software such as Tec Soft 2D Design. At first appearance this software looks basic but, depending on the skill of the designer, quite complex designs can be produced.
   The example shown is a simple block of material with initials.



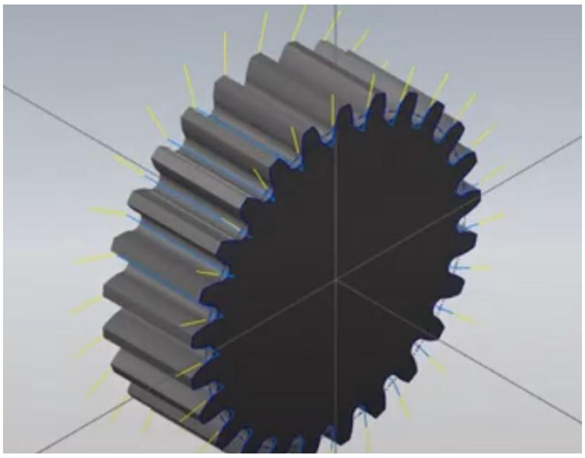

2. When the design is complete the drawing is processed. This converts the drawing into a detailed series of X, Y and Z coordinates. Processing must take place before the CNC machine can cut the design from material.

When the CNC machine shapes the material the cutter follows the coordinates, in sequence, until the shape has been manufactured

Most CAD/CAM software allows the designer to test the manufacture of his/her design on a computer rather than actually making it. This saves time and materials. Testing designs is carried out using simulation' software. When the design is run through simulation software the computer displays the manufacturing on the screen. It also checks whether or not the design can be manufactured successfully. Many designs have to be altered before they can be made by a CNC machine.



1. After all the testing and improvements to the design, it can finally be manufactured on a CNC machine.



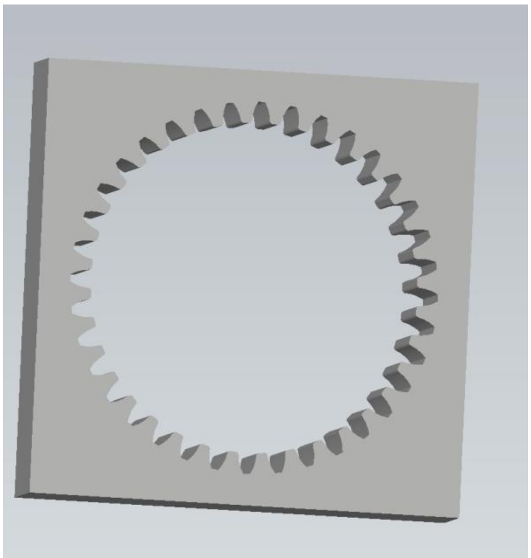
# 3D COMPUTER AIDED DESIGN AND MANUFACTURE

3D Design software allows the designer to produce three dimensional representations of his/her ideas. When completed the design can be viewed on the screen and it can even be revolved and examined at any angle. 3D software is much more complex than 2D software such as Tec Soft 2D design. It requires specialist training before it can be used competently.

- The designer draws up the design using software. The design can be examined in detailed and if modifications/alterations are needed they can be made on the screen.

Software of this type allows the designer to model his/her idea on the screen rather than make/manufacture an expensive model. Good 3D software allows the designer to design almost any item.

- The design is processed. When the design has been completed using software it must be exported as a stereo lithography file. This type of file can be imported into processing software which converts the drawing into a long list of coordinates. Each set of coordinates is called a G code & Mcode.



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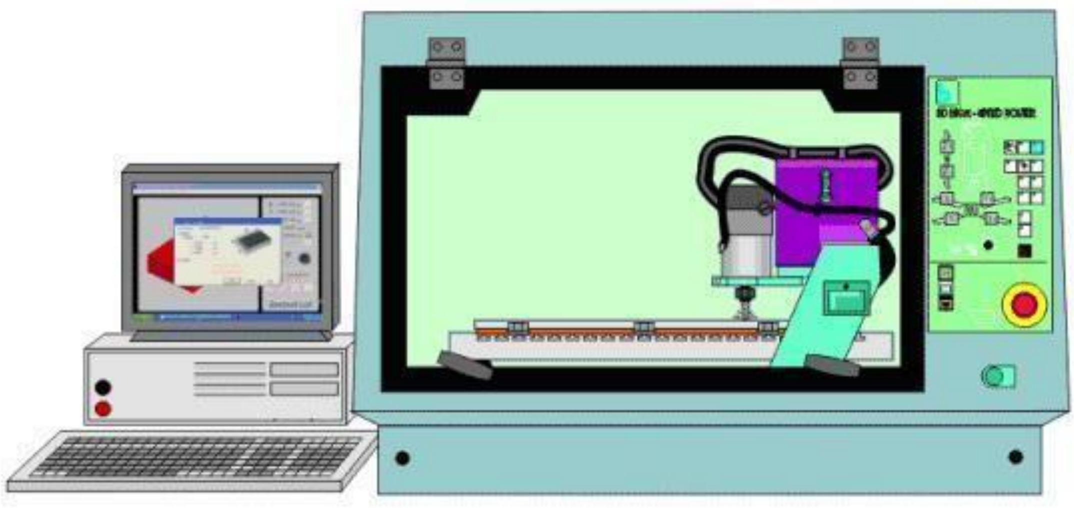
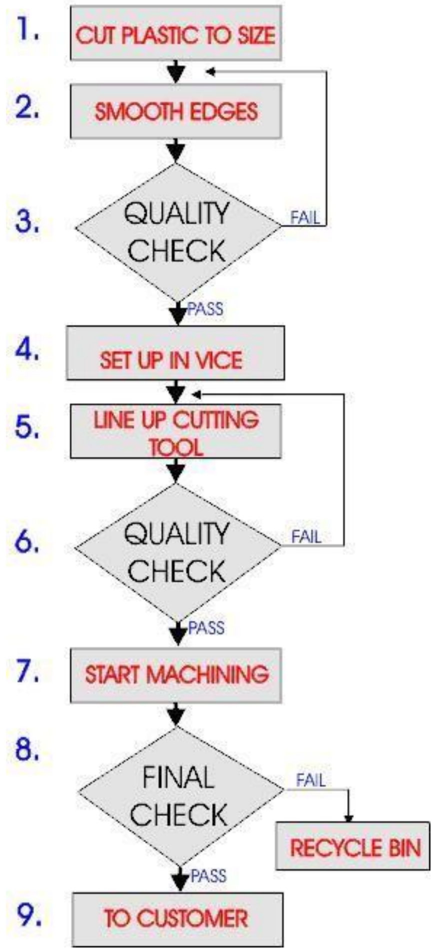
  'simulation' software. When the design is run through simulation software the computer displays the manufacturing on the screen. It also checks whether or not the design can be manufactured successfully. Many designs have to be altered before they can be made by a CNC machine.

- An advanced CNC machine can be used to manufacture the three dimensional product. This CNC is both fast and accurate making suitable for school and industrial use.



## INDUSTRIAL PRODUCTION FLOWCHART




# G & M Codes

**Mill G Codes**

G00 Rapid Positioning Motion (X,Y,Z,A,B)

G01 Linear Interpolation Motion (X ,Y,Z,A,B, F)

G02 Circular Interpolation Motion CW (X,Y,Z,A, I, J,K,R,F)

G03 Circular Interpolation Motion CCW (X,Y,Z,A, I, J,K,R,F)

G04 Dwell (P) (P=Seconds″.” Milliseconds)

G09 Exact Stop, Non-Modal

G17* Circular Motion XY Plane Selection (G02 or G03)

G18 Circular Motion ZX Plane Selection (G02 or G03)

G19 Circular Motion YZ Plane Selection (G02 or G03)

G20 Inch Coordinate Positioning

G21 Metric Coordinate Positioning

G28 Machine Zero Return Thru Ref. Point (X,Y,Z,A,B)

G29 Move to Location Through G28 Ref. Point (X ,Y,Z,A,B)

G40* Cutter Comp Cancel G41/G42/G141 (X ,Y)

G41 2D Cutter Compensation, Left (X ,Y,D)

G42 2D Cutter Compensation, Right (X ,Y,D)

G43 Tool Length Compensation + (H,Z)

G49* Tool Length Compensation Cancel G43/G44/G43

G52 Work Offset Positioning Coordinate

G53 Machine Positioning Coordinate, Non-Modal (X ,Y,Z,A,B)

G54* Work Offset Positioning Coordinate #1 (Setting 56)

G55 Work Offset Positioning Coordinate #2

G56 Work Offset Positioning Coordinate #3

G57 Work Offset Positioning Coordinate #4

G58 Work Offset Positioning Coordinate #5

G59 Work Offset Positioning Coordinate #6

G73 HS Peck Drilling Canned Cycle (X ,Y,A,B,Z, I , J ,K,Q,P,R, L, F)

G74 Reverse Tapping Canned Cycle (X ,Y,A,B,Z, J ,R, L ,F)

G76 Fine Boring Canned Cycle (X,Y,A,B,Z,I,J,P,Q,R,L,F)

G77 Black Bore Canned Cycle (X,Y,A,B,Z,I, J,Q,R,L,F)

G80* Cancel Canned Cycle (Setting 56)

G81 Drill Canned Cycle (X ,Y,A,B,Z,R, L ,F)

G82 Spot Drill / Counter bore Canned Cycle (X,Y,A,B,Z,P,R, L, F)

G83 Peck Drill Deep Hole Canned Cycle (X,Y,A,B,Z, I,J,K,Q,P,R,L,F)

G84 Tapping Canned Cycle (X,Y,A,B,Z,J,R,L,F)

G85 Bore In ~ Bore Out Canned Cycle (X ,Y,A,B,Z,R, L ,F)

G86 Bore In ~ Stop ~ Rapid Out Canned Cycle (X ,Y,A,B,Z,R, L, F)

G87 Bore In ~ Manual Retract Canned Cycle (X,Y,A,B,Z,R,L,F)

G88 Bore In ~ Dwell ~ Manual Retract Canned Cycle (X,Y,A,B,Z,P,R,L,F)

G89 Bore In ~ Dwell ~ Bore Out Canned Cycle (X,Y,A,B,Z,P,R,L,F)

G90* Absolute Positioning Command

G91 Incremental Positioning Command

G92 Global Work Coordinate System

G93 Inverse Time Feed Mode ON

G94 Inverse Time Feed OFF / Feed Per Minute ON

G98 Canned Cycle Initial Point Return

G99 Canned Cycle R Plane Return

**Mill M Codes**

M00 Program Stop

M01 Optional Program Stop

M02 Program End

M03 Spindle ON Clockwise (S)

M04 Spindle ON Counterclockwise (S))

M05 Spindle Stop

M06 Tool Change (T)

M08 Coolant ON

M09 Coolant OFF

M30 Program End and Reset

M31 Chip Auger Forward

M33 Chip Auger Stop

M34 Coolant Spigot Position Down, Increment

M35 Coolant Spigot Position Up, Decrement

M36 Pallet Part Ready

M41 Spindle Low Gear Override

M42 Spindle High Gear Override

M50 Execute Pallet Change

M83 Auto Air Jet ON

M84 Auto Air Jet OFF

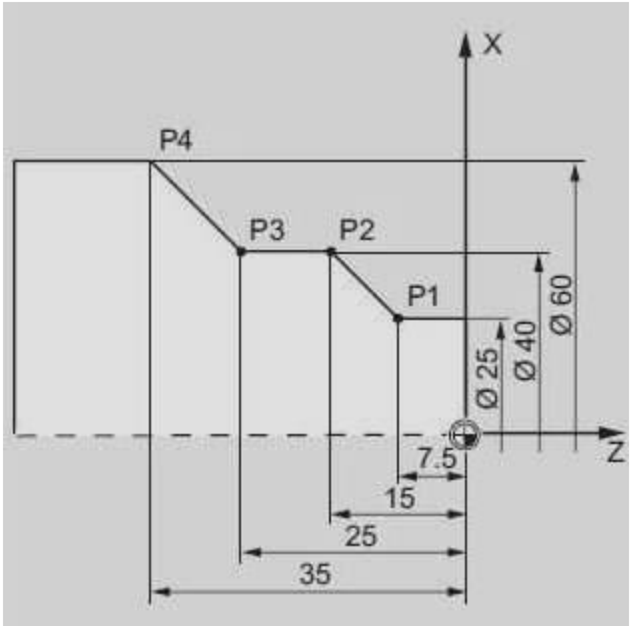
M88 Coolant Through Spindle ON

M97 Local Sub-Program Call (P, L )

M98 Sub-Program Call (P, L )

M99 Sub-Program / Routine Return of Loop (P)

**CNC Lathe Program Example with Code**




N1T0101 ; Tool no 1 with offset no 1 FANUC Control

N2G97S500M03 ; Spindle rotation clockwise with 500 RPM

N3G42G00X0Z0 ; P0 tool nose radius compensation active

N4G01X25G95F0.3 ;

N5G01Z-7.5 ; P1

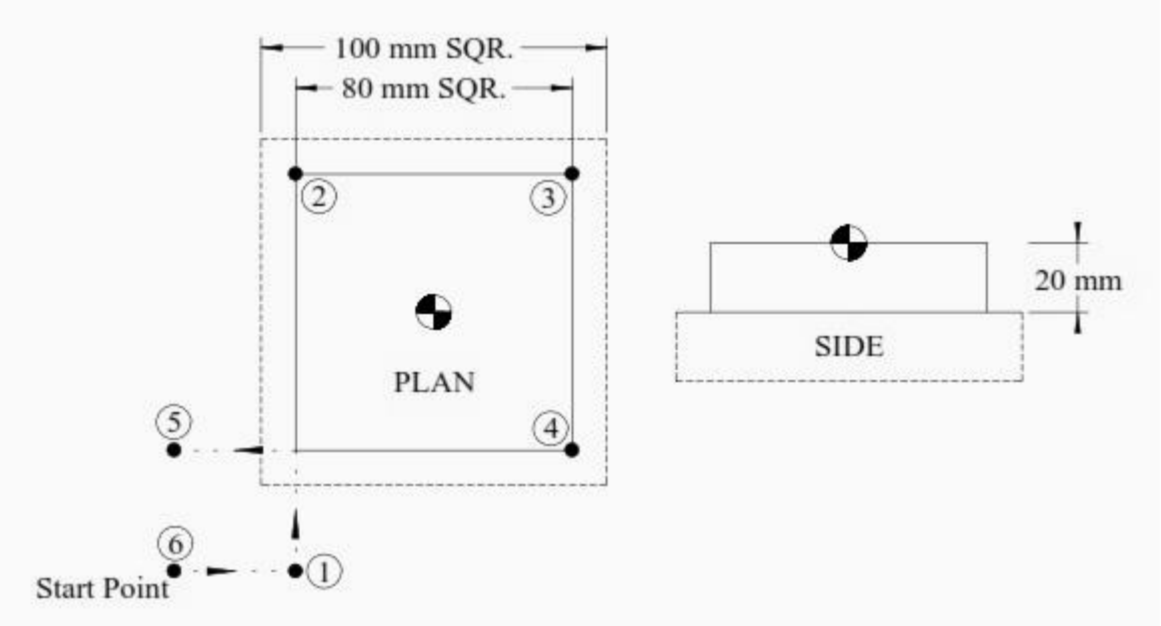
N6G01X40Z-15 ; P2

N7G01Z-25 ; P3

N8G01X60Z-35 ; P4

N9G40G00X200Z100 ; Tool nose radius compensation cancel

**Simple G Code Example Mill**




O1000

T1M6

(Linear / Feed - Absolute)

G0G90G40G21G17G94G80

G54X-75Y-75S500M3(Position 6)

G43Z100 H1

Z5

G1Z-20F100

X-40(Position 1)

Y40M8(Position 2)

X40(Position 3)

Y-40(Position 4)

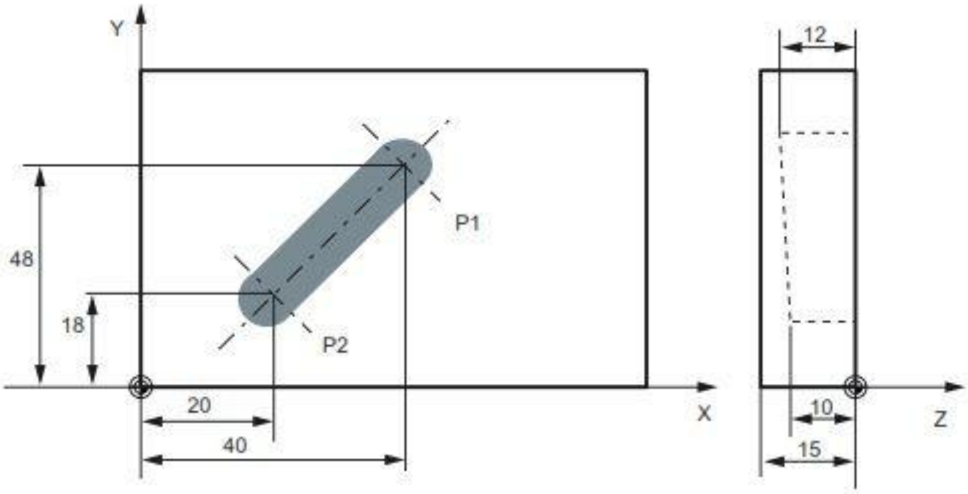
X-75(Position 5)

Y-75(Position 6)

G0Z100

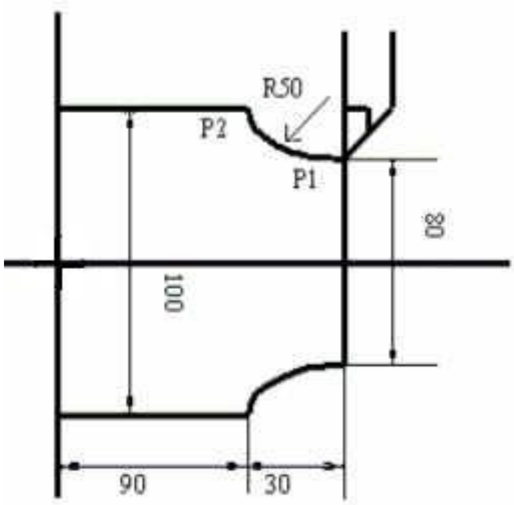
M30

### CNC Mill Program Example




```
N05G0G90X40Y48Z2S500M3
N10G1Z-12F100
N15X20Y18Z-10
N20G0Z100
N25X-20Y80
N30M2
```

### CNC Programming Example G Code G02




```
N10T0101
N20G92S1000M42
N30G96S200M03
```

```
N40G00X0Z5
N50G01Z0F0.5
N60G01X80F0.2
N70G02X100Z-30 I50 K0
N80G01Z-120
N90G00X200Z200
M30
```

**Explanation of CNC Code**

- G92 : Spindle speed limitation.
- G96 : Constant surface speed.
- G02 : Circular Interpolation Clockwise.

**G02 Explanation**

G02 has some values with it in cnc programming block like x, z, I, K.

- X : End point of arc in x-axis.
- Z : End point of arc in z-axis.
- I : Distance from arc start point to arc center point in x-axis.
- K : Distance from arc start point to arc center point in z-axis.

- **SPUR GARE PROGRAM:**

```
O0000(11111)

( T1 | | H1 )
N100 G21
N102 G0 G17 G40 G49 G80 G90
N104 T1 M6
N106 G0 G90 G54 X-70.393 Y0. A0. S4774 M3
N108 G43 H1 Z50.
N110 Z10.
N112 G1 Z-.2 F954.8
N114 G2 X-71.769 Y.196 I3.253 J27.81 F1909.6
N116 G3 X-71.802 Y.199 I-.033 J-.196
N118 X-72.001 Y0. I0. J-.199
N120 X-71.802 Y-.199 I.199 J0.
N122 X-71.769 Y-.196 I0. J.199
N124 G2 X-70.393 Y0. I4.629 J-27.614
N126 G1 X-49.775 Y49.775
N128 G2 X-50.61 Y50.887 I21.965 J17.365
N130 G3 X-50.771 Y50.969 I-.161 J-.116
N132 X-50.969 Y50.771 I0. J-.198
N134 X-50.887 Y50.61 I.198 J0.
N136 G2 X-49.775 Y49.775 I-16.253 J-22.8
N138 G1 X0. Y70.393
N140 G2 X.196 Y71.769 I27.81 J-3.253
N142 G3 X.199 Y71.802 I-.196 J.033
N144 X0. Y72.001 I-.199 J0.
N146 X-.199 Y71.802 I0. J-.199
N148 X-.196 Y71.769 I.199 J0.
N150 G2 X0. Y70.393 I-27.614 J-4.629
N152 G1 X49.775 Y49.775
N154 G2 X50.887 Y50.61 I17.365 J-21.965
N156 G3 X50.969 Y50.771 I-.116 J.161
```

- N158 X50.771 Y50.969 I-.198 J0.
- N160 X50.61 Y50.887 I0. J-.198
- N162 G2 X49.775 Y49.775 I-22.8 J16.253
- N164 G1 X70.393 Y0.
- N166 G2 X71.769 Y-.196 I-3.253 J-27.81
- N168 G3 X71.802 Y-.199 I.033 J.196
- N170 X72.001 Y0. I0. J.199
- N172 X71.802 Y.199 I-.199 J0.
- N174 X71.769 Y.196 I0. J-.199
- N176 G2 X70.393 Y0. I-4.629 J27.614
- N178 G1 X49.775 Y-49.775
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- N184 X50.969 Y-50.771 I0. J.198
- N186 X50.887 Y-50.61 I-.198 J0.
- N188 G2 X49.775 Y-49.775 I16.253 J22.8
- N190 G1 X0. Y-70.393
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- N371 X67.14 Y-23.81 I4. J0.
- N372 X67.802 Y-23.865 I0. J-4.
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- N382 X31.067 Y64.818 I0. J-24.198
- N383 G2 X27.81 Y63.14 I-3.257 J2.322
- N384 X23.81 Y67.14 I0. J4.
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- N387 M5
- N388 G91 G28 Z0.
- N389 G28 X0. Y0. A0.
- N390 M30

- **HELICAL GEAR PROGRAM:**

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  - N110 Z5.
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  - N156 X-22.074 Y9.149
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- N290 X-21.566 Y-21.566 I-36.641 J-27.81
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- N300 X27.146 Y27.146 I28.75 J27.81
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- N368 X74.969 Y-50.771 I0. J24.198
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- N387 M5
- N388 G91 G28 Z0.
- N389 G28 X0. Y0. A0.
- N390 M30

**INDUSTRY VISIT PHOTO**

## SUMMARY

- CNC programming and operations play a crucial role in modern manufacturing processes, revolutionizing the way products are designed and produced. This report has provided an overview of CNC programming, highlighting its importance, principles, and techniques. We explored the key components and functions of a CNC system, including the machine, controller, and tooling.
- Moreover, the report delved into the programming aspects of CNC, discussing the various programming languages, such as G-code and M-code, and their role in instructing the machine's movements and operations. We examined the steps involved in CNC programming, from initial design and CAD/CAM software utilization to toolpath generation and simulation.
- Furthermore, the report shed light on the operations performed using CNC machines, emphasizing their precision, repeatability, and efficiency. We explored the diverse applications of CNC technology in industries such as automotive, aerospace, electronics, and medical device manufacturing.
- Additionally, the report highlighted the benefits of CNC programming and operations, including increased productivity, improved accuracy, reduced human error, and flexibility for complex geometries and customized designs. These advantages contribute to enhanced product quality, shorter production cycles, and overall cost savings for manufacturers.
- However, it is important to acknowledge that CNC programming and operations require skilled personnel who possess a deep understanding of programming languages, machining principles, and machine maintenance. Training and ongoing education programs are crucial to developing a proficient workforce capable of maximizing the potential of CNC technology.

# Conclusion

- CNC mean computer numerical control machine it's a form of programmable automation drill drawings on wood use g coding consist of 3 motors and their drivers and pic with its basic circuit and body made of wood hold on motors and drill and the wood we want to draw on it
- We tried to make cheap fast safety CNC machine that drill on wood piece according to any drawing we draw to it.